6.1200 Quiz 1 — Review Sheet

Wed Oct 14, 7:30 PM · ℕ includes 0 · appendix = given to you on the quiz

1 · Logic

- **A → B** is false only when A true, B false. A → B ≡ ¬A ∨ B ≡ ¬B → ¬A (contrapositive). Converse B → A is **not** equivalent.
- ¬(A∧B) ≡ ¬A ∨ ¬B · ¬(A∨B) ≡ ¬A ∧ ¬B
- ¬∀x P ≡ ∃x ¬P · ¬∃x P ≡ ∀x ¬P
- **Valid** = true for every assignment. **Satisfiable** = true for at least one.
- Equivalence: truth table, or chain of known equivalences.
- Quantifier order matters: ∀x∃y ≠ ∃y∀x.

2 · Proof methods

- **Direct:** assume hypothesis → derive conclusion. **Contrapositive:** prove ¬B → ¬A.
- **Contradiction:** assume claim false → reach something impossible.
- **Cases:** split by remainder (even/odd, n mod 4); cover every case.
- **Iff:** prove both directions.
- **Well-Ordering:** assume counterexamples exist → take the smallest → build a smaller one → contradiction.

3 · Induction

- **P(n) is about one n** — no ∀n inside it, n not inside ∃.
- **Ordinary:** base P(0); step: assume P(n), prove P(n+1); conclude ∀n.
- **Strong:** "for n ≥ (last base + 1), assume P(k) for all 0 ≤ k < n; prove P(n)."
- # base cases = how far back the step reaches (uses n−2 → 2 bases).
- Stuck? Prove a **stronger** statement.

4 · State machines

- **Invariant:** true at **every** start state + preserved by every transition ⇒ by the **Invariant Principle**, true in every reachable state.
- **Unreachable:** target state breaks the invariant.
- **Finding one:** a combination of variables every move leaves unchanged (or unchanged mod m) — or a quantity that only goes up (or down).
- **Termination:** derived variable in ℕ that strictly decreases each move.
- **Partial correctness:** invariant + "it stopped" ⇒ answer correct. **Total** = partial + termination.

5 · Sums

- appendix Σk, Σk^2, finite geometric sum.
- **Memorize:** $\Sigma k^3 = (n(n+1)/2)^2$ · $\Sigma_{k\geq 0}\, x^k = 1/(1-x)$ for $|x| < 1$.
- **Start at m:** $\Sigma_{k=m}^{n} = \Sigma_{k=1}^{n} - \Sigma_{k=1}^{m-1}$. Plug in the smallest n to check!

Shape	Method
$k \cdot x^k$	Perturbation: S − xS, solve for S
polynomial in k	Ansatz: degree +1, solve coefficients, prove by induction
f(k+1) − f(k)	Telescoping: = f(n+1) − f(1)
no closed form	Integral bounds
sum in a sum	Do inner sum first

> **Integral bounds** — increasing f: $f(1) + \int_1^n f \leq \Sigma f(k) \leq f(n) + \int_1^n f$. Decreasing f: swap f(1) and f(n).
> Harmonic: $\ln n + 1/n \leq H_n \leq \ln n + 1$.

6 · Asymptotics

- appendix $f \in O(g)$: ∃c, n_0 such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Prove it by picking c and n_0.

lim f/g	< ∞	> 0	finite, > 0	0	∞	1
means	O	Ω	Θ	o	ω	~

- **Growth:** $\log n \ll n^{\varepsilon} \ll n^k \ll c^n \ll n! \ll n^n$
- appendix **Stirling:** $n! \sim \sqrt{(2\pi n)}(n/e)^n$ — for n!, binomials, log(n!) = Θ(n log n).
- ~ cares about constant factors, Θ doesn't. $4^n \notin O(2^n)$.

7 · Recurrences

> **Master Theorem** appendix $T(n) = aT(n/b) + f(n)$, $c = \log_b a$
> 1) $f \in O(n^{c-\varepsilon}) \to \Theta(n^c)$
> 2) $f \in \Theta(n^c) \to \Theta(n^c \log n)$
> 3) $f \in \Omega(n^{c+\varepsilon})$ and $a \cdot f(n/b) \leq k \cdot f(n)$, $k < 1 \to \Theta(f)$

- **ε** = the gap between f's exponent and c (any $0 < \varepsilon \leq$ gap). Write it out. Same power + extra log → no ε → theorem doesn't apply. Never for T(n−1).
- **Characteristic eq.** (a_n from previous terms, constant coefficients): $a_n \to r^2$, $a_{n-1} \to r$, $a_{n-2} \to 1$. Roots → $a_n = \lambda_1 r_1^n + \lambda_2 r_2^n$. Double root r → $(\lambda_1 + \lambda_2 n) r^n$. +constant → add a guess C. Find λ's last from starting values; check one more term.
- **Plug & chug:** unroll → k-th step → reach base case → sum → prove by induction.

8 · Divisibility & GCD

- $a \mid b \Leftrightarrow b = ak$ for some integer k. $a \mid b$ and $a \mid c \Rightarrow a \mid (sb + tc)$.
- p prime, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.
- $\gcd(a,b) = \gcd(b, \text{rem}(a,b)) = \gcd(a - b, b)$.
- **Bézout:** $\gcd(a,b) = sa + tb$. Numbers of the form $sa + tb$ are exactly the multiples of gcd.
- **Pulverizer:** run Euclid; each row $r = x - q \cdot y$; substitute earlier rows so $r = s \cdot a + t \cdot b$; stop at the gcd. Check $sa + tb = \gcd$.

9 · Modular arithmetic

- **$a \equiv_n b$** $\Leftrightarrow n \mid (a - b) \Leftrightarrow$ same remainder.
- **M1–M3:** acts like =. **M4–M7:** add/multiply congruences → swap any number for its remainder anytime.
- Reduce the base, never the exponent mod n. Cancel k only if $\gcd(k, n) = 1$.
- **Inverse** a^{-1} ($a \cdot x \equiv_n 1$) exists $\Leftrightarrow \gcd(a,n) = 1$. Pulverizer: $sa + tn = 1 \Rightarrow a^{-1} \equiv_n s$.
- appendix **FLT:** p prime, $p \nmid a \Rightarrow a^{p-1} \equiv_p 1$. So $a^{-1} \equiv_p a^{p-2}$.
- **Big powers:** reduce base; find k with $a^k \equiv_n 1$ (n prime → $k = n-1$). Every full block of k copies is 1, so $a^e \equiv_n a^{\text{rem}(e,k)}$. Otherwise square repeatedly.
- Squares $\equiv_4 0$ or 1 · odd squares $\equiv_8 1$ · $n \equiv_9$ its digit sum.

10 · Lessons from the practice quiz

1. **Plug in a small n** to check every closed form.
2. **Write ε** in the Master Theorem; check regularity in case 3.
3. Claims about all integers → **cases by remainder**, not induction. Part (b) usually uses part (a).
4. **ab odd ⇒ both a and b odd.**
5. Closing lines: "true at start, preserved ⇒ by the Invariant Principle…" · "for $n \geq 2$, assume P(k) for all $0 \leq k < n$".